

TEEP RESEARCH PROJECT B

Model Training Plan

Which models, which data goes into each, and how each one will be scored

Rachit Ranka

The data is prepared. This is the plan for the modelling phase.

What the project spec asks for

Section I of the spec lists the models, in this order.

	The spec's words	What that means in practice
1	Random Forest / Gradient Boosting on hand-crafted pulse-transit-time and PPG-morphology features	Measure things about each pulse first — width, area, timing — then fit trees to them
2	1D-CNN on raw PPG (+ECG) waveform	Give the network the signal itself and let it work out what matters
3	ResNet1D / lightweight Transformer	Deeper versions of the same idea, which is what the recent PulseDB papers use

The spec also names one thing NOT to build: it calls the leakage-style numbers from older papers “reference-only, not to be re-implemented” — they are quoted to show the inflation, not benchmarked.

And it names the method that is meant to be the contribution: “a lightweight few-shot personalization layer — a small per-subject fine-tuning of the final regression layers ... using only a handful of labeled calibration segments per new subject”.

The pipeline the spec describes

Section H, in the spec's own order.

Raw synchronised PPG + ECG + ABP segments (PulseDB)



Preprocessing — bandpass filtering, PPG/ECG fiducial point detection



Segmentation — use PulseDB's official 10-second segments



Feature / representation — pulse-transit-time and morphology features, plus a raw-waveform branch



Baseline models — trained and subject-independently validated on PulseDB splits



Proposed model — baseline plus few-shot personalization



Evaluation — (1) PulseDB subject-independent test, (2) domain-shift test on PPG-DaLiA

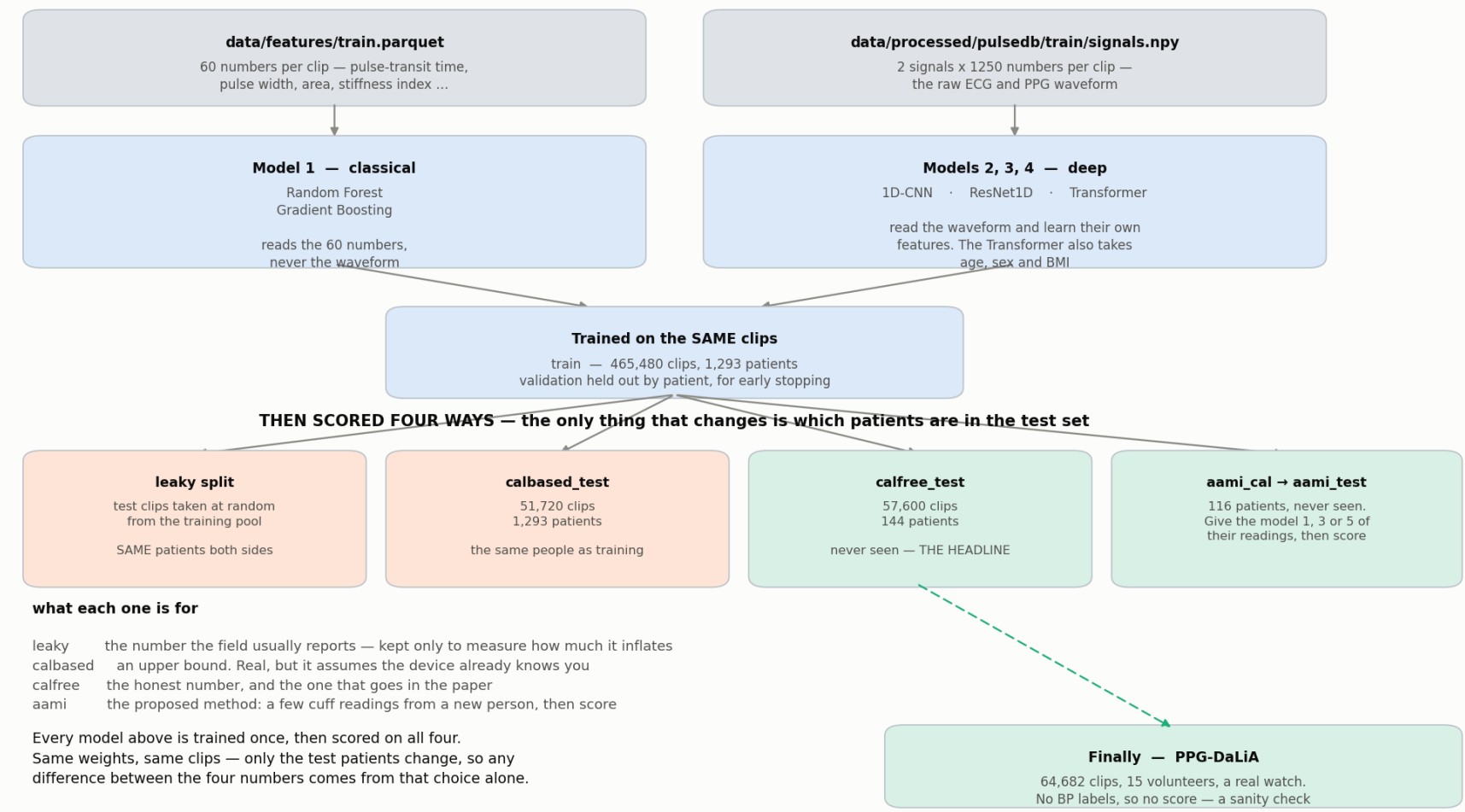
The first three steps are the preparation already reported. The rest is what follows.

Which data goes into which model

Which data goes into which model, and where each one is scored

Grey = a file on disk. Blue = a model. Orange = scored on patients it has already met. Green = scored on patients it never met.

TWO WAYS OF LOOKING AT THE SAME 465,480 CLIPS



The two ways of feeding the same clips

Both families train on the identical 465,480 clips from 1,293 patients.

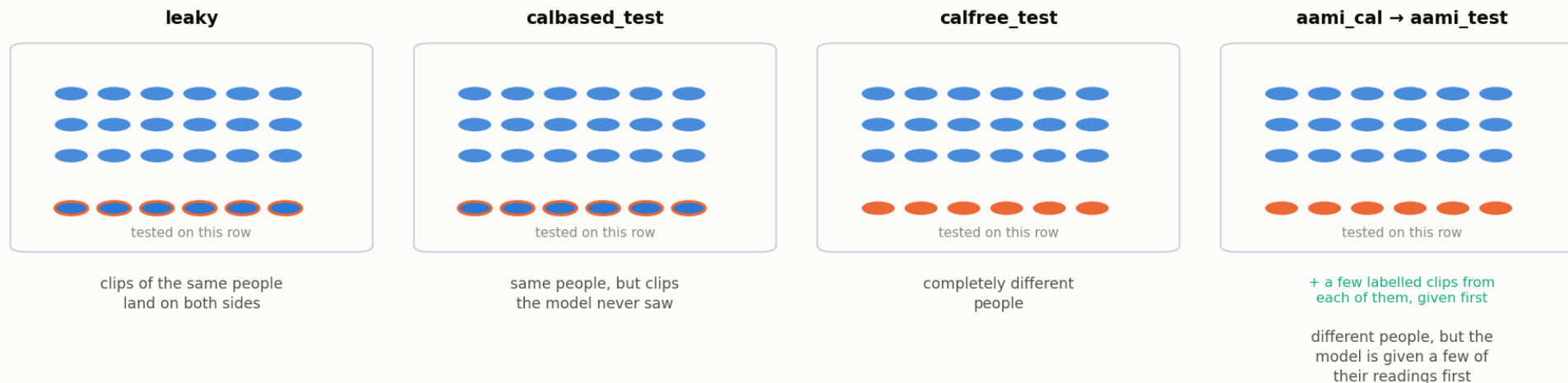
	Model 1 — classical	Models 2, 3, 4 — deep
Reads	data/features/*.parquet	data/processed/pulsedb/*/signals.npy
Per clip	60 numbers	2 signals x 1250 numbers
What those are	Pulse-transit time, pulse width, rise time, area ratio, stiffness index ...	The raw ECG and PPG waveform
Who chose them	We did — measured with a pulse detector	Nobody. The network learns its own
Extra inputs	Age, sex, height, weight, BMI	Transformer also takes age, sex, BMI
Runs on	CPU, minutes	GPU; the Transformer needs a cloud GPU

The comparison between the two is one of the spec's ablations: “classical features vs. raw-waveform CNN”. It answers whether hand-measuring the pulse is still worth doing, or whether the network does better left alone.

The four ways each model will be scored

The four ways of scoring, drawn

Each box is the pool of patients. Blue = used for training. Orange = used for testing.



In the first two the test people are already in the training rows — the model has met them. In the last two it has not. The gap between the first and the third is what this project is built to measure. The fourth is the fix it proposes.

The first two score the model on people it has already met. The last two do not.

The gap between them is what the project exists to measure, and the fourth is the fix it proposes.

How the training runs will be set up

Section L of the spec, point by point.

What the spec asks for	How it will be done
Subject-level split	Use PulseDB's official groups rather than inventing our own
Cross-validation	k-fold by patient inside the training group, for tuning only
Prevention of leakage	Assertions in code — no patient ID shared between train, calibration and test
Target imbalance	BP is skewed in ICU patients — stratified sampling or a weighted loss across BP bins
Personalization protocol	For a held-out patient, use only the first few of their clips as calibration; score on the rest

The spec calls the leakage assertions “itself worth reporting as a methodological contribution, since it directly documents what much prior work skipped”.

How the results will be measured

Section K. Two metrics are primary because the clinical standard needs both.

Metric	Why	Role
Mean absolute error (MAE)	The number the whole field reports, for SBP and DBP	Primary
Standard deviation of error	AAMI requires both the mean error and its spread	Primary
AAMI / BHS compliance	Mean error ≤ 5 mmHg and SD ≤ 8 mmHg — the clinical bar	Primary
Pearson correlation	Does the prediction move with the truth at all	Secondary
Bland-Altman plots	The standard way of comparing two BP methods	Secondary

A good average error alone is not enough. A model can be right on average and still swing wildly from reading to reading, which is why the spec makes the spread primary as well.

The method the spec calls the contribution

Everything above is a baseline. This is the part that is meant to be new.

A new patient puts on the watch. The model has never seen them, and on its own it is not accurate enough to be useful.

So we ask them for a handful of real cuff readings — once, at home or at a clinic visit — and use those to adjust the model to that person. Only the final regression layers are re-fitted, or a simple shift is applied in feature space. Nothing is retrained from scratch.

The question that follows is the one nobody has answered:

how few readings are enough? 1? 3? 5?

That is a planned ablation in the spec, and it is the difference between a device an elderly person can set up at home and one they cannot.

The experiments this produces

Experiment	What it compares	What it answers
Leakage gap	Same model, leaky split vs patient-separated split	How much does the wrong split inflate?
Model family	Classical features vs raw-waveform networks	Is hand-measuring the pulse still worth it?
Architecture	CNN vs ResNet vs Transformer, all patient-separated	Does a deeper model actually help here?
Personalization	Calibration-free vs 1, 3, 5 calibration clips	How few readings does a new person need?
Domain shift	Hospital-trained model run on watch data	Does any of it survive outside the ICU?

Each row is a comparison, not a single number. The comparisons are the result.

What other groups reported on this same dataset

All of these use PulseDB, and all use its official groups.

Study	Year	Model	Patients it had seen	Brand new patients
PulseDB (the dataset paper)	2023	1D ResNet-18	not reported	not reported
Validation study	2024	CNN + LSTM + attention	not reported	13.67 / 8.56
Validation study	2024	UTransBPNNet (U-Net + Transformer)	not reported	12.50 / 8.32
Benchmark study	2025	XResNet1d101 (best of five)	9.08 / 6.08	12.70 / 8.05
DMT	2026	Transformer + age, sex, BMI	4.56 / 2.62	12.17 / 7.89
rU-Net	2024	U-Net + ResNet + attention	4.49 / 2.69	4.14 / 2.48 after tuning

All figures are mean absolute error in mmHg, written SBP / DBP. Lower is better.

The last row is the exception that matters: rU-Net's second number comes after giving the model 10% of the new patient's own readings.

The pattern in those numbers

Read the last two columns of the previous slide across, not down.

Patients the model had already seen about 4 to 9 mmHg

Patients it had never seen about 12 to 14 mmHg

Never seen, but given a few of their own readings about 4 mmHg

Three things follow from that.

One. The gap between the first two lines is not one paper's weakness. Every study that reports both shows it. It is the honest difficulty of the problem.

Two. Four years of new architectures moved the middle line by about 1.5 mmHg in total — from 13.67 in 2024 to 12.17 in 2026. The model is not the bottleneck.

Three. The third line is a different order of improvement, and it comes from giving the model a little information about the person rather than from a better network.

What we should expect, and what we should aim for

Setting	Realistic expectation	Why
Leaky split	roughly 5 to 8 mmHg	Deliberately inflated — reported only to measure it
Patients already seen	roughly 5 to 9 mmHg	Matches what every paper reports
Brand new patients	roughly 12 to 13 mmHg	Where the whole field sits, regardless of model
After personalization	roughly 4 to 5 mmHg	The one published precedent reports 4.14 / 2.48

Building a deeper network is worth doing, but it is worth roughly 1 mmHg. Personalization is worth roughly 8. The plan is weighted accordingly.

The spec agrees: it says the project is a result even “without beating any state-of-the-art number”, provided the leakage gap, the personalization recovery and the domain shift are all quantified honestly.

References

TEEP Research Project Proposals — Project B, sections H, I, J, K, L.

PulseDB: a large, cleaned dataset based on MIMIC-III and VitalDB — Wang et al.,
Frontiers in Digital Health, 2023. doi.org/10.3389/fdgth.2022.1090854

Generalizable deep learning for PPG-based blood pressure estimation — a benchmarking study, 2025.
[arXiv:2502.19167](https://arxiv.org/abs/2502.19167)

DMT: Demographic Conditioning, Morphology-Enhanced Transformer, 2026. [arXiv:2606.11125](https://arxiv.org/abs/2606.11125)

Validation of deep learning models for cuffless BP estimation on a large benchmarking dataset, 2024.
oaepublish.com/articles/chatmed.2023.23

rU-Net, Multi-Scale Feature Fusion and Transfer Learning, IEEE JBHI, 2024.
ieeexplore.ieee.org/document/10721372

Investigation of Data Leakage in Deep-Learning-Based Blood Pressure Estimation, IEEE, 2023.